

Abhi Mondal
2RMS075

PH4207 : Q1P
Problem Set II

Answer: $|\psi_{AB}\rangle = \frac{1+\sqrt{6}}{2\sqrt{6}}|00\rangle + \frac{1-\sqrt{6}}{2\sqrt{6}}|01\rangle + \frac{\sqrt{2}-\sqrt{3}}{2\sqrt{6}}|10\rangle + \frac{\sqrt{2}+\sqrt{3}}{2\sqrt{6}}|11\rangle$

$$\begin{aligned} \rho_{AB} &= |\psi_{AB}\rangle\langle\psi_{AB}| \\ &= (a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle)(a\langle 00| + b\langle 01| + c\langle 10| + d\langle 11|) \\ &= (a^2|00\rangle\langle 00| + ab|00\rangle\langle 01| + ac|00\rangle\langle 10| + ad|00\rangle\langle 11| + \\ &\quad ab|01\rangle\langle 00| + b^2|01\rangle\langle 01| + bc|01\rangle\langle 10| + bd|01\rangle\langle 11| + \\ &\quad ac|10\rangle\langle 00| + bc|10\rangle\langle 01| + c^2|10\rangle\langle 10| + cd|10\rangle\langle 11| + \\ &\quad ad|11\rangle\langle 00| + bd|11\rangle\langle 01| + cd|11\rangle\langle 10| + d^2|11\rangle\langle 11|) \end{aligned}$$

$$\rho_{AB} = \begin{pmatrix} a^2 & ab & ac & ad \\ ab & b^2 & bc & bd \\ ac & bc & c^2 & cd \\ ad & bd & cd & d^2 \end{pmatrix}$$

Note that here, $a = \frac{1+\sqrt{6}}{2\sqrt{6}}$; $b = \frac{1-\sqrt{6}}{2\sqrt{6}}$; $c = \frac{\sqrt{2}-\sqrt{3}}{2\sqrt{6}}$; $d = \frac{\sqrt{2}+\sqrt{3}}{2\sqrt{6}}$

To verify the purity of this state,

First Note that, $\text{Tr}(\rho_{AB}) = a^2 + b^2 + c^2 + d^2$

$$\text{Tr}(\rho_{AB}) = \frac{1}{4.6} [2(1+6) + 2(2+3)] = \frac{1}{24} \times (14+10) = 1$$

$$\begin{aligned} \text{Next, } \text{Tr}(\rho_{AB}^2) &= \text{Tr}(|\psi_{AB}\rangle\langle\psi_{AB}| \underbrace{|\psi_{AB}\rangle\langle\psi_{AB}|}_{\text{II}}) \\ &= \text{Tr}(|\psi_{AB}\rangle\langle\psi_{AB}|) \\ &= 1 \end{aligned}$$

So, Also by construction, $|\psi_{AB}\rangle$ is pure state, and by $\text{Tr}(\rho_{AB}^2) = 1$ it is pure state.

$$\rho_A = \text{Tr}_B |\psi_{AB}\rangle\langle\psi_{AB}|$$

$$\begin{aligned} &= (a^2|0\rangle\langle 0| + ac|0\rangle\langle 1| + c^2|1\rangle\langle 0| + b^2|0\rangle\langle 0| + bd|0\rangle\langle 1| + \\ &\quad + bd|1\rangle\langle 0| + d^2|1\rangle\langle 1|) \\ &= (a^2+b^2)|0\rangle\langle 0| + (ac+bd)|0\rangle\langle 1| + (ac+bd)|1\rangle\langle 0| + (c^2+d^2)|1\rangle\langle 1| \end{aligned}$$

$$a^2+b^2 = \frac{(1+\sqrt{6})^2}{(2\sqrt{6})^2} + \frac{(1-\sqrt{6})^2}{(2\sqrt{6})^2} = 2 \times (1+6) = 14$$

$$ac+bd = \frac{(1+\sqrt{6})(\sqrt{2}-\sqrt{3})}{(2\sqrt{6})^2} + \frac{(1-\sqrt{6})(\sqrt{2}+\sqrt{3})}{(2\sqrt{6})^2}$$

$$= \frac{\sqrt{2}-\sqrt{3} + \sqrt{3}-3\sqrt{2} + \sqrt{2}+\sqrt{3} - 3\sqrt{2}-3\sqrt{3}}{24} = -\frac{4\sqrt{2}}{24}$$

$$c^2 + d^2 = (\sqrt{2} - \sqrt{3})^2 + (\sqrt{2} + \sqrt{3})^2$$

$$= 2 \times (2+3) = 10$$

$$P_A = \frac{1}{4 \cdot 6} \begin{pmatrix} 14 & -4\sqrt{2} \\ -4\sqrt{2} & 10 \end{pmatrix} = \begin{pmatrix} 14/24 & -4\sqrt{2}/24 \\ -4\sqrt{2}/24 & 10/24 \end{pmatrix}$$

$$P_A = \begin{pmatrix} 7/12 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 5/12 \end{pmatrix}$$

$$P_B = a^2 |0\rangle\langle 0| + ab |0\rangle\langle 1| + ab |1\rangle\langle 0| + b^2 |1\rangle\langle 1| + c^2 |0\rangle\langle 0| + cd |0\rangle\langle 1| +$$

$$cd |1\rangle\langle 0| + d^2 |1\rangle\langle 1|$$

$$= (a^2 + c^2) |0\rangle\langle 0| + (ab + cd) |0\rangle\langle 1| + (ab + cd) |1\rangle\langle 0| + (b^2 + d^2) |1\rangle\langle 1|$$

$$a^2 + c^2 = \frac{(1+\sqrt{6})^2 + (\sqrt{2}-\sqrt{3})^2}{4 \cdot 6} = \frac{1+6+2\sqrt{6}+2+3-2\sqrt{6}}{4 \cdot 6} = \frac{12}{24} = \frac{1}{2}$$

$$ab + cd = \frac{(1+\sqrt{6})(1-\sqrt{6}) + (\sqrt{2}-\sqrt{3})(\sqrt{2}+\sqrt{3})}{4 \cdot 6}$$

$$= \frac{1-6+2-3}{24} = \frac{-6}{24} = -1/4$$

$$b^2 + d^2 = \frac{(1-\sqrt{6})^2 + (\sqrt{2}+\sqrt{3})^2}{24} = \frac{1+6-2\sqrt{6}+2+3+2\sqrt{6}}{24}$$

$$= \frac{12}{24} = \frac{1}{2}$$

$$P_B = \begin{pmatrix} 1/2 & -1/4 \\ -1/4 & 1/2 \end{pmatrix}$$

3. Finding Eigenvalues of P_A and P_B

$$\det(P_A - \lambda I) \rightarrow \begin{vmatrix} 7/12 - \lambda & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 5/12 - \lambda \end{vmatrix}$$

$$\det(P_A - \lambda I) = (7/12 - \lambda)(5/12 - \lambda) - \frac{2}{3818} = 0$$

$$\Rightarrow \cancel{35} \lambda^2 - \frac{7}{12} \lambda - \frac{5}{12} \lambda + \frac{35}{144} - \frac{1}{18} = 0$$

$$\Rightarrow \lambda^2 - \lambda + \frac{35-8}{144} = 0 \Rightarrow \lambda^2 - \lambda + \frac{27}{144} = 0$$

$$\Rightarrow \lambda = \frac{1 \pm \sqrt{1 - \frac{4 \times 27}{144}}}{2} = \frac{1 \pm \sqrt{1 - 3/4}}{2}$$

$$= \frac{1 \pm 1/2}{2}$$

$$= \frac{1}{2} \pm \frac{1}{4}$$

$$\lambda_1 = 3/4$$

$$\lambda_2 = 1/4$$

$$\frac{4 \times 27}{144}$$

$$= \frac{3}{4}$$

do Finding eigen vectors.

$$\lambda_1 = 3/4 \quad \begin{pmatrix} 7/12 - 3/4 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 5/12 - 3/4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 7-9 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 5-9 & 12 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -1/6 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & -4/3 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow -\frac{u_1}{6} - \frac{\sqrt{2}u_2}{6} = 0 \rightarrow u_1 = -\sqrt{2}u_2$$

$$-\sqrt{2}u_1 = 2u_2 \rightarrow u_1 = -\sqrt{2}u_2$$

$$v_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{2} \\ -1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{2} \\ -1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{3}} (\sqrt{2}107 - 117)$$

$$\lambda_2 = 1/4 \quad \begin{pmatrix} 7/12 - 1/4 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 5/12 - 1/4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 7-3 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 5-3 & 12 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4/12 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 2/12 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1/3 & -\sqrt{2}/6 \\ -\sqrt{2}/6 & 1/6 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \frac{u_1}{3} - \frac{\sqrt{2}u_2}{6} = 0 \rightarrow 2u_1 - \sqrt{2}u_2 = 0 \rightarrow \sqrt{2}u_1 = u_2$$

$$\Rightarrow -\sqrt{2}u_1 + u_2 = 0 \rightarrow \sqrt{2}u_1 = u_2$$

$$\text{So } v_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{3}} (107 + \sqrt{2}117)$$

• Finding Eigenvalues P_B

$$\det(P_B - \lambda I) = \begin{vmatrix} 1/2 - \lambda & -1/4 \\ -1/4 & 1/2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (1/2 - \lambda)^2 - 1/16 = 0 \Rightarrow \lambda = 1/2 \pm 1/4 \Rightarrow \lambda_1 = 3/4 \text{ \& } \lambda_2 = 1/4$$

Finding eigenvectors of P_B .

$$\begin{pmatrix} 1/2 - 3/4 & -1/4 \\ -1/4 & 1/2 - 3/4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} \bar{V}_1 \rangle_B = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \bar{V}_2 \rangle_B = \frac{1}{\sqrt{2}} (10\rangle - 11\rangle) \end{cases}$$

$$\begin{pmatrix} 1/2 - 1/4 & -1/4 \\ -1/4 & 1/2 - 1/4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} \bar{V}_2 \rangle_B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ \bar{V}_1 \rangle_B = \frac{1}{\sqrt{2}} (10\rangle + 11\rangle) \end{cases}$$

So, the Schmidt decomposition form of $|\psi^{AB}\rangle$ is,

$$|\psi^{AB}\rangle = \frac{\sqrt{3}}{2} |\bar{V}_1\rangle_A |\bar{V}_1\rangle_B + \frac{1}{2} |\bar{V}_2\rangle_A |\bar{V}_2\rangle_B$$

$$\begin{aligned} |\bar{V}_1\rangle_A &= \frac{1}{\sqrt{3}} (\sqrt{2}10\rangle - 11\rangle) \\ |\bar{V}_2\rangle_A &= \frac{1}{\sqrt{3}} (10\rangle + \sqrt{2}11\rangle) \end{aligned} \quad \left| \begin{aligned} |\bar{V}_1\rangle_B &= \frac{1}{\sqrt{2}} (10\rangle - 11\rangle) \\ |\bar{V}_2\rangle_B &= \frac{1}{\sqrt{2}} (10\rangle + 11\rangle) \end{aligned} \right.$$

Ans. 2

$$|\psi^{AB}\rangle = \frac{1}{2} [10\rangle_A 10\rangle_B - 10\rangle_A 11\rangle_B + 11\rangle_A 10\rangle_B + 11\rangle_A 11\rangle_B]$$

$$= \frac{1}{2} [100\rangle - 101\rangle + 110\rangle + 111\rangle]$$

$$\beta = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\beta^\dagger \beta = \frac{1}{2} \times \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\beta^\dagger \beta = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

So eigenvalues are $1/2$ and $1/2$

$$\sigma_1 = \frac{1}{\sqrt{2}}, \sigma_2 = \frac{1}{\sqrt{2}}$$

So, the Diagonal matrix is $\Sigma = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

So, the eigenvectors of $\beta^\dagger \beta$ is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$\text{So } V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow V^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Now by singular value decomposition,

$$\beta = U \Sigma V^T$$

$$\rho = U \Sigma V^T$$

$$= U \left(\frac{1}{2} I \right) U^T$$

$$= \frac{1}{2} U$$

$$\text{So } U = \sqrt{2} \rho$$

$$\Rightarrow U = \sqrt{2} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\boxed{\rho = U \Sigma V^T}$$

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} ; U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\Sigma = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Finding state of entanglement

$$\lambda_1 = \frac{1}{2}, \lambda_2 = \frac{1}{2} \text{ So,}$$

we can easily find Von Neuman Entropy,

$$S = - \left[\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right] = - \left[\frac{1}{2} (-1) + \frac{1}{2} (-1) \right] = 1$$

So the state is maximally disordered or entangled state.